Syllabus for

Academic year 2020/2021 >

TIF275 - Tools of engineering physics

Fysikingenjörens verktyg

Syllabus adopted 2020-02-10 by Head of Programme (or corresponding)
Owner: TKTFY
10,5 Credits
Grading: UG - Pass, Fail
Education cycle: First-cycle
Major subject: Engineering Physics
Department: 16 - PHYSICS

Teaching language: Swedish
Application code: 57133
Open for exchange students: No
Only students with the course round in the programme plan

Credit distribution

Module				Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0113	Examples class	1,5 c	Grading: UG	1,5 c						
0213	Laboratory	1,5 c	Grading: UG	1,5 c						
0313	Project	1,5 c	Grading: UG	1,5 c						
0413	Project	3,0 c	Grading: UG		3,0 c					
0513	Project	3,0 c	Grading: UG			3,0 c				

In programs

TKTFY ENGINEERING PHYSICS, Year 1 (compulsory)

Examiner:

Jonathan Weidow

Go to course homepage

Theme:

Environment 1,5 hec

Eligibility

General entry requirements for bachelor's level (first cycle)
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Eligibility for the programme.

Aim

The purpose of this introductory course is to equip the students with basic skills in handling some of the common tools employed by physicists and engineers. This comprises both concrete tools relevant for their continued education, e.g. handling the local computer systems and mathematical software, but also experimental methodology and basic principles of physical modeling and analysis. In addition to this, the course serves to orient the students in some of the contemporary research areas where physicist are working today. Communication is an integrated part of the curriculum for the course. The course aims at developing basic genre consciousness along with writing characterized by correctness and feeling for stylistic differences with regard to recipient. In the last part of the course the focus of the communication part shifts to oral presentation in the context of project presentations. The introduction of a sustainability perspective in the analysis of phenomena in physics aims at developing a high sense of sustainability in the future engineers. It is also to be an introduction to the course Environmental Physics. The course also aims to enable students to start developing entrepreneurial skills.

Learning outcomes *(after completion of the course the student should be able to)*

* Use the most common UNIX- and LaTeX-commands.
* Find commands and flags in UNIX and LaTeX using available resources.
* Solve more complex tasks requiring a combination of several UNIX-commands.
* Use basic functionality in MATLAB as a tool for numerical computation and visualization in mathematics and physics.
* Write text, characterized by correctness in language and style, oriented towards a given recipient.
* Work with process oriented writing.
* Use a sustainability perspective in an analysis of physical physical phenomena.
* Plan and perform experimental investigations of physical phenomena and analyze the results.
* Convert labnotes to a proper report.
* Perform basic dimensional analysis and physical modelling.
* Treat measurement uncertainties.

* Receive and provide constructive feedback on written and oral presentations.
* Perform basic CAD.
* Perform basic 3D printing.
* Plan, structure and give an oral presentation.
* Apply an entrepreneurial approach to the analysis and modeling of physical phenomena.

Content

The part 0113 (1,5 hec) treats:
* UNIX: Shell, file management, permissions, paths, processes, editing, the help system and a small amount of programming. * LaTeX: Formatting, disposition, tables, figures, mathematical typesetting, references, help resources and common errors.

The part 0213 (1,5 hec) treats:
* Desktop layout and the help functionality.
* Arithmetic, assignments, variables and standard built in functions.
* Script files and function files. Graphics.
* Finding zeros of a function numerically.
* Solving linear systems of equations.
* Numerical quadrature and solving of differential equations.
* Basic programming and program structures.
* Handling of data files.

The part 0313 (1.5 hec) treats:
* Order of magnitude estimation, units and dimensional analysis.
* Physical modelling.
* Technical writing.
* Sustainable development in different physics areas.
* Entrepreneurship in physics.

The part 0413 (3.0 hec) treats:
* Experimental methodology (planning, execution, documentation, analysis and treatment of errors).
* Physical modelling.
* Interpretation of measurement data and curve fitting.
* Writing a labreport.

The part 0513 (3.0 hec) treats:
* The physics and mathematics behind toys designed to illustrate physical and mathematical phenomena.

* Free-form prototype production (3D printing). * 3D modelling with Computer-aided design (CAD). * Presentation techniques and training in oral presentation with a particular focus on a physical phenomenon.

Organisation

The parts 0113 and 0213 are based on lectures and computer exercises. The parts 0313, 0413 and 0513 are project based and includes lectures, exercises, lab-classes, workshops and seminars.

Examination including compulsory elements

0113 and 0213 are assessed by demonstrating skill in solving tasks at the computer.

Pass grade in 0313, 0413 and 0513 requires: active participation in compulsory seminars, labs and workshops. Project presentations, oral and written, must comply with both textual as well as technical standards. Students must participate in giving feedback to other students on their projects. Hand-in assignments and/or additional midterm may comprise part of the assessment.